Nematollah (Emad) Saeidi

AI/ML Engineer and Researcher

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Summary

- AI/ML engineer and researcher working across applied machine learning, computer vision, generative AI, and medical image analysis. Experience spans local RAG voice assistants, speaker and face recognition, visual search, recommendation systems, customer analytics, and research on graph neural networks and foundation models for medical image retrieval.

Experience

Hyrcan Technology Development Co., AI Engineer — Tehran, Iran (Hybrid)
Develop local generative AI and real-time identity recognition applications. — Nov 2025 – present — 1 year

- Built a local RAG-based voice assistant with LangChain, Ollama, JSON/ChromaDB, and Sentence Transformers.
- Applied few-shot prompting and conversation memory to support contextual voice dialogue.
- Integrated Whisper speech-to-text, RAG-enhanced LLM generation, and Coqui text-to-speech into a complete speech pipeline.
- Implemented real-time speaker recognition with NVIDIA NeMo TitaNet and Resemblyzer, incorporating voice activity detection and acoustic feature extraction.
- Implemented real-time face recognition pipelines using YOLO for detection and ArcFace ResNet100 / InsightFace for recognition.

Danube Private University Austria (DPU), MIAAI Group, AI Researcher — TFZ, Austria (Hybrid)
Research medical image retrieval and histopathology image analysis in the Medical Image Analysis & Artificial Intelligence (MIAAI) group. — Sept 2023 – present — 3 years 2 months

- Investigate graph neural networks and foundation models for content-based medical image retrieval.
- Work on histopathological image classification using nuclei-based graph convolutional networks and foundation-model features.
- Apply image classification, nuclei segmentation, and generative modeling methods to medical image analysis.

FANAP Co., Marketing Data Scientist — Tehran, Iran (Remote)
Supported AI service evaluation and business development while contributing to applied research projects. — July 2021 – Sept 2023, 2 years 3 months

- Collaborated with cross-functional teams to troubleshoot and optimize more than 10 deployed AI services.
- Designed and executed evaluations of model accuracy, speed, and reliability; identified and reported anomalies in AI systems.
- Drafted requests for proposals (RFPs) and business plans for initiatives in the AI open innovation ecosystem.
- Authored technical articles and research briefs for the FANAP Research Team.
- Medical image search engine (Dec 2022–Sep 2023): worked on retrieval and classification using attention-based adversarial variational graph autoencoders, pretrained models, and approximate nearest-neighbor search.
- Drawing-robot motion planning (Mar–Jul 2022): developed image-processing methods to extract contours for a drawing robot.
- Worked within an agile process using daily standups, sprint retrospectives, backlog grooming, and continuous integration in two-week sprints.

FANAP Co., Data Scientist — Tehran, Iran (Hybrid)
Developed computer vision, recommendation, and customer analytics applications at FANAP. — Oct 2018 – July 2021, 2 years 10 months

- Object detection API (Sep 2020–Jul 2021): implemented detection and segmentation workflows using few-shot models and transfer learning.
- Video surveillance attendance API (Jan–Mar 2021): implemented an attendance and registration application.
- Automatic number plate recognition (Jan 2021): implemented vehicle registration, vehicle-data search, and ranked record display by license plate.
- Visual search API (Jan–Sep 2020): combined object detection and recognition, color extraction, image crawling, transfer learning, and approximate nearest-neighbor search.
- Customer analytics and clustering library (Aug 2019–Jan 2020): developed SaaS metrics and KPIs, segmentation, and cluster ensembles for numeric and categorical grocery-retail data.
- Transaction analysis (Aug 2019–Jan 2020): clustered customer behavior in credit-card transactions to support fraud detection, including money-laundering and tax-evasion analysis.
- Recommendation system (Mar–Aug 2019): developed sequence-to-sequence ranking, association rules, hybrid/content-based/collaborative filtering, customer clustering, and feature engineering for candidate generation.
- Marketing analysis and product prediction (Oct 2018–Feb 2019): applied CatBoost, XGBoost, LightGBM, ensemble methods, and multilayer perceptrons across business datasets.
- Collaborated on fraud detection, chatbot, video surveillance, recommendation, and customer behavior analysis services.
- Worked in two-week agile sprints with daily standups, retrospectives, backlog grooming, and continuous integration.

University of Isfahan, Research Assistant — Isfahan, Iran (On-site)
Conducted research on content-based medical image retrieval using graph neural networks. — Oct 2019 – July 2023, 3 years 10 months

- Worked with graph neural networks, generative models, and foundation models for medical image analysis and retrieval.

Education

PhD **University of Isfahan**, Artificial Intelligence — Isfahan, Iran
Thesis: Content-based Medical Image Retrieval Using Graph Neural Networks. — Jan 2019 – Oct 2026

- Thesis: Content-based Medical Image Retrieval Using Graph Neural Networks.

MSc **University of Isfahan**, Artificial Intelligence — Isfahan, Iran
Thesis: Learning Multi-Objective Binary Features for Image Representation and Recognition. — Jan 2015 – Jan 2017
- Thesis: Learning Multi-Objective Binary Features for Image Representation and Recognition.

BSc **Islamic Azad University Najaf Abad**, Software Engineering — Isfahan, Iran
Thesis: Intelligent Agent-based Social Simulation Model using Reinforcement Learning. — Jan 2010 – Jan 2013
- Thesis: Intelligent Agent-based Social Simulation Model using Reinforcement Learning.

Publications

Unfolding Graph Convolutional Networks: Architectures, Use Cases, and Trends — Jan 2026
Nematollah Saeidi, Vahid Hajipour, S. Hekmat, B. Shoushtarian
10.1016/j.asoc.2026.115122 (Applied Soft Computing (IF~8.2, Q1))

Leveraging Medical Foundation Model Features in Graph Neural Network-Based Retrieval of Breast Histopathology Images — Jan 2026
Nematollah Saeidi, H. Karshenas, B. Shoushtarian, S. Hatamikia, R. Woitek, Amirreza Mahbod
10.1002/ima.70346 (International Journal of Imaging Systems and Technology (IF~3.3, Q2))

Evaluating Pre-trained Convolutional Neural Networks and Foundation Models as Feature Extractors for Content-based Medical Image Retrieval — Jan 2025
Amirreza Mahbod, Nematollah Saeidi, S. Hatamikia, R. Woitek
10.1016/j.engappai.2025.110571 (Engineering Applications of Artificial Intelligence (IF~8, Q1))

Integrating Foundation Model Features into Graph Neural Network and Fusing Predictions with Standard Fine-Tuned Models for Histology Image Classification — Jan 2025
Nematollah Saeidi, N. Torbati, R. Woitek, Amirreza Mahbod
10.3390/bioengineering12121332 (Bioengineering, 12, 1332 (IF~3.7, Q2))

Incorporating Shape Dependent Power Law in Motion Planning for Drawing Robots — Jan 2024
M. Abedinzadeh Shahri, Nematollah Saeidi, Vahid Hajipour
10.1016/j.robot.2024.104801 (Robotics and Autonomous Systems, 182, 104801 (IF~4.3, Q1))

Artificial Intelligence and Clinical Decision Making: Approaches and Challenges — Jan 2022
Nematollah Saeidi, M. Torabi
10.22034/JAISIS.2022.371239.1054 (Journal of Applied Intelligent Systems and Information Sciences, 3(2), 43-55)

Attention-Based and Positional-Aware Neural Networks for Next-Item Recommendation — Jan 2021
Nematollah Saeidi, H. Shahamat
10.22034/JAISIS.2021.304225.1033 (Journal of Applied Intelligent Systems and Information Sciences, 2(1), 29-40)

Single Sample Face Recognition Using Multicross Pattern and Learning Discriminative Binary Features — Jan 2019
Nematollah Saeidi, H. Karshenas, H. M. Mohammadi
10.1080/19361610.2019.1581877 (Journal of Applied Security Research, 14(2), 169-190 (IF~1.2, Q2))

Learning Multi-Objective Binary Features for Image Representation Jan 2017
Nematollah Saeidi, H. Karshenas, H. M. Mohammadi
10.1109/ICCKE.2017.8167927 (2017 7th International Conference on Computer and Knowledge Engineering (ICCKE), Mashhad, pp. 47-52, IEEE)

Awards

Ernst Mach Grant, OeAD (Austrian Agency for International Cooperation in Education and Research) Jan 2024
Research stay at Medical Image Analysis & Artificial Intelligence (MIAAI), Danube Private University (DPU), Austria.
OeAD

Ernst Mach Grant, OeAD (Austrian Agency for International Cooperation in Education and Research) Jan 2023
Research stay at Medical Image Analysis & Artificial Intelligence (MIAAI), Danube Private University (DPU), Austria.
OeAD

Third place in the call to solve business challenges Jan 2021
FANAP Co.

Second in the graduating class, MSc in Computer Engineering (Artificial Intelligence) Jan 2017
University of Isfahan

Top two percent of graduated students, ranked 3rd among 158 BSc graduates in Computer Engineering Jan 2013
Islamic Azad University Najaf Abad

Top Student, scholarships for two semesters Jan 2011
Islamic Azad University Najaf Abad

Reviewer

- Applied Soft Computing, Elsevier (Jul 2026 – present)
- Computer Methods and Programs in Biomedicine, Elsevier (Jun 2026 – present)
- Biomedical Signal Processing and Control, Elsevier (Sep 2024 – present)
- International Journal of Imaging Systems and Technology, Wiley (Sep 2026 – present)
- Cancer Control, SAGE
- Journal of Applied Intelligent Systems & Information Sciences (JAISIS), FANAP (May 2021 – present)

Skills

Generative AI and LLMs
Generative AI, GenAI models, Large Language Models (LLM), Local LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Fine Tuning, Finetuning Large Language Models, Pretraining LLMs, Hugging Face, Transformers, Vector indexing, Voice Chat, LangChain, Ollama, Sentence Transformers, Whisper, Coqui TTS, ChromaDB

Medical Image Analysis
Medical Image Analysis, Histopathology image analysis, Digital Pathology, Foundation Models, Graph Neural Networks, Image Retrieval, Image classification

Computer Vision
Computer Vision, OpenCV, Detectron2, Digital Image Processing, Image Analysis, Image Search, Facial Recognition, Speaker Recognition, Transfer Learning, Approximate Nearest Neighbor Search, ImageAI, Darknet, YOLO, ArcFace, InsightFace, NVIDIA NeMo TitaNet, Resemblyzer

Machine Learning and Deep Learning
Machine Learning, Deep Learning, Neural Networks, PyTorch, TensorFlow, Keras, Gradient Boosting (CatBoost, LightGBM, XGBoost), Recommender Systems, Artificial Intelligence, Scikit-learn

Programming and Tools
Python, Matlab, NetLogo, Git, LaTeX, Flask, RESTful WebServices, Postman, Programming, Pytest, Flask-RESTful, Docker, PyCharm, Jupyter, Anaconda, Linux, Windows

Data
Data Science, Data Analysis, Data Analytics, Data Visualization, Tableau, DASK, SQL, Elastic-search, NumPy, Pandas, Matplotlib, SQL Server, Faiss

Research and Professional
Research, Research and Development (R&D), Agile Methodologies, Microsoft Office, Word-Press

LinkedIn Skill Assessments passed
Python, Machine Learning, Matlab, Agile Methodologies, Git

Web Development
Bootstrap, CSS, JavaScript, WordPress

Projects

Medical Image Retrieval with Foundation Models
Evaluation of pre-trained CNNs and medical foundation models as feature extractors for content-based medical image retrieval.

Histology Image Classification with GNNs
Medical image classification using foundation model features in a graph neural network.

Photo-Sketch
Contour generation from input images for drawing-robot motion planning.

Content-Based Medical Image Retrieval across 2D and 3D Images
Investigated vision and vision-language foundation models for content-based retrieval across 2D and 3D medical images, including the effect of image size on retrieval performance.

Automatic Number Plate Recognition
Vehicle registration, vehicle-data search, and retrieval of records based on license plates.

Chatbot User Interface
Developed a chatbot user interface.

Fraud Detection and Transaction Analysis
Analyzed transaction data and customer behavior to identify anomalies and potential fraud risks.

Video Surveillance Attendance and Registration
Implemented an API-based attendance and registration application using video surveillance.

Object Detection Application
Developed object detection and segmentation workflows using few-shot models and transfer learning, exposed through a REST API.

Visual Search Application
Combined object detection and recognition, color extraction, image crawling, transfer learning, and approximate nearest-neighbor search.

Customer Behavior Analysis
Built customer segmentation and clustering workflows, including SaaS metrics, KPIs, cluster ensembles, and numeric and categorical clustering for grocery retail.

Recommendation System
Developed sequence-to-sequence ranking, association rules, hybrid/content-based/collaborative filtering, customer clustering, and candidate-generation features.

Marketing Analysis and Product Prediction
Applied CatBoost, XGBoost, LightGBM, ensemble methods, and multilayer perceptrons to marketing and prediction datasets.

Multi-Objective Binary Features for Image Representation and Recognition
Developed MATLAB implementations for robust binary feature extraction for face recognition and image representation.

Sugarscape Simulation with Reinforcement Learning
Developed an agent-based social simulation model using reinforcement learning as part of undergraduate research.

Certificates

Pretraining LLMs — Apr 2025
DeepLearning.AI

Finetuning Large Language Models — Mar 2025
DeepLearning.AI

Applied Plotting, Charting and Data Representation in Python — Oct 2022
University of Michigan (Coursera)

Python Project for AI and Application Development — Sept 2022
IBM

Applied Machine Learning in Python — May 2022
University of Michigan (Coursera)

Fifth IPM Advanced School on Computing and Artificial Intelligence — Sept 2021
Institute for Research in Fundamental Sciences (IPM)

Fourth IPM Advanced School on Computing and Artificial Intelligence — Aug 2020
Institute for Research in Fundamental Sciences (IPM)

IPM Advanced Data Science Summer School (ADS3) — Aug 2018
Institute for Research in Fundamental Sciences (IPM)

IPM Elementary Data Science Summer School (EDS3) Jun–Jul 2018
Institute for Research in Fundamental Sciences (IPM)

Teaching Assistant

- **Machine Learning** — Graduate Teaching Assistant, University of Isfahan, Iran (Jan 2016–Jun 2017), under the supervision of Dr. Peyman Adibi.
- **Statistical Pattern Recognition** — Graduate Teaching Assistant, University of Isfahan, Iran (Jan 2016–Jun 2017), under the supervision of Dr. Peyman Adibi.
- **Algorithm Design** — Graduate Teaching Assistant, University of Isfahan, Iran (Jan 2016–Jun 2017), under the supervision of Dr. Peyman Adibi.

Languages

- **Persian:** Native speaker | **English:** C1 | **German:** A2

Interests

Medical Image Analysis
Content-based Image Retrieval, Histopathology, Foundation Models

Graph Neural Networks
Graph Convolutional Networks, Graph Autoencoders, Generative Models

Applied AI
RAG, Speech Pipelines, Face and Speaker Recognition, Recommender Systems

References

Amirreza Mahbod
Assistant Professor, Medical Image Analysis & Artificial Intelligence (MIAAI), Danube Private University (DPU), Vienna, Austria. amirreza.mahbod@dp-uni.ac.at

Vahid Hajipour
Assistant Professor, Industrial Engineering, Altinbas University, Istanbul, Turkey; Head of Research Center, FANAP Co., Tehran, Iran. vahid.hajipour@altinbas.edu.tr